

Exploratory Data Analysis of Lenskart's E-Commerce Database

Chapter - 1

1.1 Introduction

Lenskart is one of India's leading omnichannel eyewear retailers, offering prescription glasses, sunglasses, contact lenses, and related accessories through its website, mobile app, and physical stores. Since its founding in 2010, the company has grown rapidly by combining online convenience with in-store try-on experiences, home eye check-ups, and a wide catalogue of frames spanning multiple price points and styles.

A dataset built around Lenskart's operations typically captures information such as product details (frame type, brand, price, lens type), customer demographics, order and transaction history, ratings and reviews, delivery timelines, and store or channel-wise sales. Analysing this kind of data allows us to understand customer buying behaviour, product performance, seasonal demand patterns, and areas where the business can improve its operations or customer experience.

This project uses the Lenskart dataset to explore these dimensions through data cleaning, exploratory data analysis (EDA), and visualization, with the goal of uncovering actionable business insights.

1.2 Problem Statement

Lenskart operates in a highly competitive and fast-growing eyewear market, generating large volumes of data related to customer orders, product catalogues, pricing, and reviews. However, without proper analysis, this data remains underutilized, making it difficult for the business to identify which products perform best, understand customer preferences, detect seasonal demand shifts, or pinpoint gaps in customer satisfaction.

There is a need to analyse Lenskart's dataset systematically to extract meaningful patterns and insights that can support better decision-making around inventory management, marketing strategy, pricing, and customer experience. This project aims to address that gap by applying data analysis techniques to uncover actionable insights from the available data.

1.3 Objectives

1. To understand the structure and key variables present in the Lenskart dataset.

2. To clean and preprocess the data by handling missing values, duplicates, and inconsistencies.
3. To analyse customer purchasing patterns, including preferred product categories, price ranges, and order frequency.
4. To identify top-performing products, brands, or categories based on sales volume and revenue.
5. To study customer satisfaction through ratings, reviews, or return/complaint data (if available).
6. To visualize trends over time, such as monthly or seasonal variations in demand.
7. To generate insights that could support business decisions around inventory, marketing, or customer retention.

1.4 Motive/ Purpose

The main purpose of this project is to analyse Lenskart's retail data and gain practical experience in data analytics. It helps us understand customer behaviour, identify sales trends, and generate insights that can support better business decisions. The project also allows us to apply data cleaning, analysis, and visualization techniques to a real-world dataset while developing valuable analytical skills.

1.5 Stakeholders

1. **Business Owners/Management** – Interested in overall sales performance, profitability, and strategic decision-making.
2. **Marketing Team** – Uses customer behaviour insights to design targeted campaigns and promotions.
3. **Product/Inventory Team** – Needs insights on best-selling products, categories, and demand trends to manage stock levels.
4. **Customer Service Team** – Uses satisfaction and review data to improve customer experience and resolve recurring complaints.
5. **Data Analysts/Data Science Team** – Responsible for performing the analysis and generating reports/dashboards.
6. **Customers** (indirect stakeholders) – Benefit from improved products, services, and pricing driven by data-backed decisions.
7. **Investors/Shareholders** – Interested in business growth trends and performance metrics derived from data insights.

1.6 Business Requirement

1. The analysis should provide a clear understanding of sales performance across products, categories, and time periods.
2. The system should identify customer buying patterns and preferences to support marketing and personalization strategies.
3. The analysis should highlight top and low-performing products/brands to guide inventory decisions.
4. The project should surface customer satisfaction trends through ratings/reviews (if available in the dataset).
5. Insights should be presented in a clear, visual, and easily interpretable format (charts, graphs, dashboards) for non-technical stakeholders.
6. The analysis should be reproducible and adaptable for future/updated datasets.

1.7 Functional Requirements

1. **Data Import** – Ability to load the Lenskart dataset (CSV/Excel) into the analysis environment.
2. **Data Cleaning** – Ability to handle missing values, duplicate records, and inconsistent formatting.
3. **Data Exploration** – Ability to generate summary statistics (mean, median, mode, distribution) for key variables.
4. **Visualization** – Ability to create charts (bar, line, pie, heatmaps, etc.) to represent trends and comparisons.
5. **Segmentation/Grouping** – Ability to group data by category, brand, region, or time period for comparative analysis.
6. **Insight Generation** – Ability to derive and document key findings from the analysis.
7. **Reporting** – Ability to export results/insights into a report or presentation format.

1.8 Non-Functional Requirements

1. **Performance** – The analysis process should handle the dataset efficiently without excessive processing time.

2. Usability – Visualizations and reports should be easy to understand for both technical and non-technical stakeholders.
 3. Scalability – The approach should be adaptable to larger or updated versions of the dataset in the future.
 4. Accuracy – Data cleaning and analysis methods should ensure minimal errors and reliable insights.
 5. Reliability – The analysis should produce consistent results when re-run on the same dataset.
 6. Maintainability – Code and analysis scripts should be well-documented and easy to modify or extend.
 7. Security/Privacy – If the dataset contains any customer-sensitive information, it should be handled in compliance with data privacy practices.
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